CAMPUSX 100 DAYS OF MACHINE LEARNING

DAY 1

Complete Beginner Notes & Practice Workbook

Three videos covered

- What is Machine Learning?
- AI vs ML vs DL for Beginners in Hindi
- Types of Machine Learning for Beginners

Definitions • Roman Urdu explanations • comparison tables • worked examples • Python practice • exercises • answer key • mini-project • revision checklist

Daily rule: study at least one hour and always finish a video you start, even if it takes longer.

How to use this workbook

Review the concepts after watching the official videos. Explain each idea in your own words, run the Python example, and attempt the exercises before checking the answer key.

Official video sequence

#	Video	Official access
1	What is Machine Learning? | 100 Days of Machine Learning	Open Video 1
2	AI Vs ML Vs DL for Beginners in Hindi	Playlist — Video 2
3	Types of Machine Learning for Beginners | Types of ML in Depth	Playlist — Video 3

Use the official playlist to confirm titles and order if an individual link changes.

Learning outcomes

- Define ML and explain how it learns patterns from data.
- Identify features (X), targets (y), algorithms and trained models.
- Describe the ML workflow and unseen-data evaluation.
- Explain AI, ML and DL and their relationship.
- Distinguish regression, classification and four learning settings.
- Run and understand a basic scikit-learn model.

1. What is Machine Learning?

Machine Learning (ML) is a field within Artificial Intelligence where algorithms learn patterns from data and use those patterns to make predictions, classifications, recommendations or decisions.

Roman Urdu: Computer ko har prediction ka rule manually dene ke bajaye, hum usay examples aur data dete hain. Algorithm patterns seekhta hai aur trained model naye examples par prediction karta hai.

House-price example

House size	Illustrative price
500 sq. ft.	Rs. 2.0 million
700 sq. ft.	Rs. 2.8 million
900 sq. ft.	Rs. 3.6 million

A simple model may learn that larger houses tend to cost more and estimate Rs. 4.0 million for a 1,000 sq. ft. house in this toy dataset. This is not real market data; actual prices depend on location, condition, age and other factors.

Traditional programming vs ML

Traditional programming	Machine Learning
A human writes explicit rules; data plus rules produce an output.	An algorithm learns a model from examples; the model predicts on new inputs.
Example: written rule checks whether a number is even.	Example: learn a relationship between house size and price.

ML still requires code, data preparation, human choices and evaluation. It does not mean computers learn without guidance.

2. Essential vocabulary

Term	Meaning	Example
Data	Examples used for learning or evaluation.	Historical house records.
Feature (X)	Input information used to make a prediction.	Size, location, rooms.
Target/label (y)	Answer a supervised model aims to predict.	House price.
Algorithm	Learning procedure used to fit patterns.	Linear regression method.
Model	Fitted result of training, used on new inputs.	Trained price predictor.
Training	Fitting a model to examples.	Learning from known prices.
Prediction	Output for a given input.	Estimated price of a new house.

If study hours are used to predict exam marks, study hours are feature X and marks are target y. Roman Urdu: feature input hota hai; target woh answer hai jo predict karna hai.

3. Machine-learning workflow

- Collect data: gather relevant examples.
- Prepare data: check missing values, errors, formats and quality.
- Choose an approach: match the method to the task and data.
- Train: fit the algorithm to training data.
- Evaluate: measure performance on examples not used for fitting.
- Predict: apply the trained model to new inputs.
- Monitor: check performance over time and retrain when justified.

Why test on unseen data?

A model may memorise training examples or learn patterns that do not generalise. Evaluation on held-out data estimates how well it may work on new examples. Testing does not automatically improve a model; it provides evidence to guide decisions. Keep test data separate from training and tuning to avoid data leakage.

4. AI vs ML vs DL

Term	Meaning	Example
Artificial Intelligence (AI)	Broad field of systems performing tasks associated with intelligent behaviour.	Rule-based expert system or chess search.
Machine Learning (ML)	Subset of AI that learns patterns from data.	Predict house prices from past records.
Deep Learning (DL)	Subset of ML using neural networks with multiple layers.	Image or speech recognition.

Relationship: DL is within ML, and ML is within AI. AI can also use written rules or search without ML.

Roman Urdu: AI bara field hai; ML us ka hissa hai; DL, ML ka hissa hai. Har AI system ML use nahi karta, aur har ML problem ko DL ki zaroorat nahi hoti.

Cat recognition

A deep-learning classifier can learn visual patterns from labelled photos of cats and other objects, then predict whether a new image contains a cat. It is not guaranteed to be correct, and deep learning is not the only image-recognition approach. DL can be useful for complex image, audio and language tasks but may need more data and computing resources.

5. Types of Machine Learning

5.1 Supervised learning

Learns from examples where the correct answer (label/target) is known.

- Regression: predicts a number, such as house price, exam marks or temperature.
- Classification: predicts a category, such as spam/not spam or cat/dog.

5.2 Unsupervised learning

Finds useful structure in data without supplied target labels.

- Clustering: groups similar examples, such as customers or study-habit profiles.
- Dimensionality reduction: represents data with fewer variables while retaining useful information.

5.3 Semi-supervised learning

Combines a smaller labelled dataset with a larger unlabelled dataset. It may help when labelling is expensive, provided data and assumptions are suitable.

5.4 Reinforcement learning

An agent acts in an environment and receives rewards or penalties. It learns a policy—a strategy for choosing actions—to improve longer-term return. Rewards are task-defined feedback, not moral judgements.

Type	Learning signal	Example
Supervised	Labelled examples	Predict exam marks.
Unsupervised	Unlabelled data; discover structure	Group similar students.
Semi-supervised	Some labelled and more unlabelled data	Plant photos with few labels.
Reinforcement	Actions and reward feedback	Learn to play a game.

Two additional distinctions

- Batch vs online learning: batch learning trains from a collected dataset, often with periodic retraining; online learning updates incrementally as new examples arrive.
- Instance-based vs model-based: instance-based methods compare new inputs with stored examples; model-based methods learn a general model.

These are different dimensions for describing systems, not replacements for the four learning settings.

6. Worked examples

Study hours and marks

Practice data: (2 hours, 40 marks), (4, 55), (6, 70), (8, 85).

- Feature X = study hours.
- Target y = final marks.

- Predicting a numeric mark is regression.
- This simplified pattern does not prove study hours alone determine real exam performance.

Spam detection

If labelled emails are used to learn spam/not-spam categories, this is supervised classification. Features may include message text and sender information.

Grouping students

If group labels are not supplied and the aim is to discover groups by attendance or study habits, clustering is an unsupervised approach.

7. Python practical: linear regression

Install once: `python -m pip install scikit-learn`. These invented examples are for learning, not real-world decisions.

```
from sklearn.linear_model import LinearRegression

# Feature X: house size in square feet
X = [[500], [700], [900], [1100], [1300]]

# Target y: illustrative price in millions of rupees
y = [2.0, 2.8, 3.6, 4.4, 5.2]

model = LinearRegression()
model.fit(X, y)

prediction = model.predict([[1000]])
print(f"Predicted price: Rs. {prediction[0]:.2f} million")
```

Expected output: Predicted price: Rs. 4.00 million

Line-by-line explanation

- Import: brings the LinearRegression method into the program.
- X: input features; each inner list is one example. scikit-learn expects X as a 2D structure.
- y: target values matching rows in X.
- LinearRegression(): creates the model object.
- model.fit(X, y): fits the model to training examples.
- model.predict([[1000]]): predicts for one new example with one feature.
- .2f: formats the number to two decimal places.

8. Exercises

Exercise 1 — Identify X and y

Predict crop yield using rainfall and fertiliser amount. Identify the features and target.

Exercise 2 — Regression or classification?

(a) Predict a student's marks. (b) Predict whether a transaction is fraudulent. (c) Estimate tomorrow's temperature.

Exercise 3 — Select the learning setting

(a) Learn from labelled cat/dog photos. (b) Group customers without labels. (c) Use 200 labelled and 10,000 unlabelled photos. (d) Train a game agent using rewards.

Exercise 4 — Correct the statement

“All AI systems use ML, and every ML task needs DL.”

Exercise 5 — Unseen data

Why should a model be evaluated on examples not used to fit it?

9. Worked answer key

Exercise 1

Features X = rainfall and fertiliser amount. Target y = crop yield.

Exercise 2

(a) Regression if marks are numeric. (b) Classification if the target is fraudulent/not fraudulent. (c) Regression if temperature is predicted numerically.

Exercise 3

(a) Supervised. (b) Unsupervised, often clustering. (c) Semi-supervised. (d) Reinforcement learning.

Exercise 4

AI is broader than ML; some AI systems use rules or search without ML. DL is one type of ML and is not necessary for every ML task.

Exercise 5

Unseen-data evaluation estimates generalisation to new examples. It does not automatically improve the model; it informs decisions about whether and how to improve it.

10. Mini-project: study-hours predictor

Goal: practise the ML workflow with a small invented dataset.

- Ask: can study hours help estimate marks?
- Create practice pairs of study hours and marks; label the data as invented.
- Train a LinearRegression model using the same structure as the house-price example.
- Predict for a new study-hour value within the range of your practice data.
- Change one training example and observe how the prediction changes.
- Reflect on why study hours alone may not predict real exam performance accurately.
- Do not claim real-world accuracy without suitable data and evaluation.

11. Common beginner mistakes

- Confusing features (inputs) with targets (answers to predict).
- Treating the algorithm and trained model as identical.
- Assuming unseen-data evaluation automatically improves a model.
- Assuming every AI system uses ML.
- Assuming deep learning is always better than simpler methods.
- Confusing regression (numeric target) with classification (category).
- Treating rewards as moral approval rather than task-defined feedback.
- Allowing test data to influence training or tuning, causing data leakage.

12. Day 1 revision sheet

Question	Remember
What is ML?	Learn patterns from data to make predictions or decisions.
Feature X?	Input information used by the model.
Target y?	Output a supervised model aims to predict.
Algorithm vs model?	Learning procedure vs fitted result.
Why unseen data?	Estimate performance on new examples.
AI vs ML vs DL?	DL is within ML; ML is within AI.
Regression?	Predict a number.
Classification?	Predict a category.
Unsupervised?	Find structure without supplied target labels.
Reinforcement?	Learn a policy using actions and rewards.

13. Revision checklist

- ☐ I can explain ML in my own words.
- ☐ I can identify features X and target y.
- ☐ I can explain algorithm vs model.
- ☐ I understand unseen-data evaluation.
- ☐ I can explain the AI, ML and DL relationship.
- ☐ I can distinguish regression and classification.
- ☐ I can describe the four common learning settings.
- ☐ I can run and explain the Python example.
- ☐ I have reviewed my Day 1 assessment mistakes.

Journey rule: aim for at least one hour daily. Always finish any video you start, even if it takes longer than one hour. Review and correct misunderstandings before proceeding to the next day.